

Alex Graham.

Senior Software Developer & Systems Architect

LOCATION

North East England

WEB

a-graham.uk

GITHUB

[alexgrahamuk666](https://github.com/alexgrahamuk666)

PROJECTS

ingenious-solutions.co.uk

PROFILE

Senior software developer and systems architect with **20+ years of professional experience** across commercial agencies, higher education and freelance practice. Specialist in complex PHP-based web applications, multi-tenant SaaS architecture, legacy system modernisation and DevOps implementation. Proven track record of leading development teams, mentoring junior developers and delivering **mission-critical systems under pressure** for a variety of diverse clients. **BookN**, a scale-ready multi-client booking platform. Equally comfortable at the architecture whiteboard, the command line or the soldering iron.

EXPERIENCE

Freelance Senior Developer & Founder

2023 – Present

Self-employed / BookN / Ingenious Solutions – Sunderland

Designing and building **BookN** – a multi-client, multi-tenant booking platform built on Silverstripe Framework. Architecture pushes the boundaries of the framework's ORM and CMS to support a single API endpoint serving multiple client databases. Scale-ready from day one, informed by hard lessons from previous high-traffic production incidents.

Concurrent freelance engagements across complex PHP/MySQL backend systems, API integrations and frontend development. Operating under the **ingenious-solutions.co.uk** brand – *yet another ingenious solution*.

Analyst / Developer → Senior Analyst / Developer

2018 – 2023

Newcastle University – Newcastle upon Tyne

Promoted from Analyst/Developer to Senior Analyst/Developer. Primary custodian of **NESS** – a mission-critical academic management system handling PEC submissions, exam board review sessions, mark entry, module mark computation, stage results and final degree classifications for the entire university. System required absolute reliability during assessment periods with zero tolerance for error.

Substantially rewrote and modernised the legacy codebase; built the majority of integrations with external university systems including **SAP** and multiple third-party platforms using Swagger-documented, standards-compliant APIs.

Led the modernisation of the development team's infrastructure: introduced **Docker** containerisation, **GitHub** version control workflows and **TeamCity** CI/CD pipelines, automating many previously manual release and testing tasks. Also applied skills from earlier career to architect the team's physical infrastructure – desktop builds, servers, networking, IP PBX.

Freelance Developer

2015 – 2018

Self-employed – North East England

Complex PHP/MySQL backend and frontend systems for a range of clients. Projects included bespoke CMS builds, e-commerce integrations, API development and statistical data processing packages. Concurrent with and leading into the Newcastle University engagement.

Lead Developer

2013 – 2015

Daykin & Storey – Baltic Business Quarter, Gateshead

Continued as Lead Developer following the merger of Azure Graphic & Web Design (Sunderland) with Virtuoso (Durham) to form Daykin & Storey. Relocated to the Baltic Business Quarter offices in Gateshead. Led the development team through the transition and continued delivery of complex client projects across the combined agency portfolio.

Developer → Lead Developer
Azure Graphic & Web Design – Sunderland

2007 – 2013

Joined as Web Developer and progressed to **Lead Developer**, taking responsibility for the agency's technical output, development standards and a growing team of junior developers. Mentored junior staff – several of whom went on to senior roles and founding their own successful companies.

Managed the full stack across numerous client projects. Also applied broader technical skills to office infrastructure including desktop and server builds, network architecture, PBX systems and structured cabling – skills carried forward from the electrical trade.

Electrician's Mate
Electrical Contractor – North East England

2006 – 2007

Domestic, commercial and industrial electrical maintenance, rewires and new installations. Also involved in **PBX telephone system** installations and structured network cabling – a direct bridge between the electrical trade and the technology career that followed. Practical skills from this period have been applied throughout subsequent roles in infrastructure and systems work.

Freelance Web Developer
Self-employed – Sunderland

2006 – 2006

Post-university freelance work including uScript statistical packages, web development and early PHP projects. Established foundational commercial development practice before moving into the electrical trade.

EDUCATION

2002 – 2005
Bachelor of Science — Computing
University of Sunderland

BSc Computing. Foundation in software engineering, systems architecture and applied computing.

1999 – 2002
A-Levels • AS-Levels • Various Additional Courses
City of Sunderland College

3 A-Levels • 2 AS-Levels • DipHE Computing

1994 – 1999
GCSEs
Secondary School – Sunderland

Multiple GCSEs including Maths, English and Sciences.

SKILLS

Languages & Frameworks	Infrastructure & DevOps	Architecture & Integration
- PHP 5-8 (expert) - Silverstripe Framework & CMS - JavaScript / ES6+ - HTML5 / CSS3 - SQL / MySQL - Bash / Shell scripting	- Docker / containerisation - GitHub / Git workflows - TeamCity CI/CD - Linux (Debian, Ubuntu, RHEL) - VMware ESXi / vSphere - Network architecture - IP PBX systems	- Multi-tenant SaaS design - REST API design (Swagger/OpenAPI) - SAP integration - Legacy system modernisation - IoT systems & Raspberry Pi - Audio DSP / BPM analysis - RAID / NAS / storage
Leadership & Process	Physical & Electrical	Other
- Development team leadership - Junior developer mentoring - Agile / iterative delivery - Code review & standards - Project architecture	- Server & desktop builds - Structured network cabling - PBX installation - Domestic & commercial electrical - Hardware fault diagnosis	- Music production / audio DSP - Technical writing - Open source contribution - Statistical software (uScript) - Cloudflare / DNS / hosting